Results

RT_Repeated Measures ANOVA

Within Subjects Effects

Cases	Sphericity Correction	Sum of Squares	df	Mean Square	F	p	η^2
Condition	None	6589	2.000	3294	11.14	< .001	0.5820
	Greenhouse-Geisser	6589	1.365	4826	11.14	.004	0.5820
Residuals	None	4732	16.00	295.7			
	Greenhouse-Geisser	4732	10.92	433.3			

Note. Type III Sum of Squares

Between Subjects Effects

Cases	Sum of Squares	df	Mean Square	F	p
Residuals	$1.122\times10^{+4}$	8	1403		

Note. Type III Sum of Squares

Descriptives

Descriptives

Condition	N	Mean	SD	SE	Coefficient of variation
Simple	9	201.4	22.42	7.475	0.1113
Go/No-Go	9	214.1	30.12	10.04	0.1407
Go/Stop	9	239.0	24.17	8.057	0.1011

Assumption Checks

Test of Sphericity

	Mauchly's W	Approx. X^2	df	p-value	Greenhouse-Geisser ε	Huynh-Feldt ε	Lower Bound ε
Condition	0.5349	4.379	2	.112	0.6826	0.7752	0.5000

Nonparametrics

Friedman Test

Factor	X^2_F	df	p	Kendall's W
Condition	14.22	2	< .001	0.7901

Post Hoc Tests

Post Hoc Comparisons - Condition

		Mean Difference	SE	df	t	Cohen's d	p_{Holm}
Simple	Go/No-Go	−12.68	6.232	8	−2.035	−0.4919	.090
	Go/Stop	−37.61	6.936	8	−5.422	−1.459	.002
Go/No-Go	Go/Stop	−24.92	10.50	8	−2.374	−0.9667	.090

Note. P-value adjusted for comparing a family of 3 estimates.

MT_Repeated Measures ANOVA

Within Subjects Effects

Cases	Sphericity Correction	Sum of Squares	df	Mean Square	F	p	η^2
Condition	None	$1.501\times10^{+5}$	3.000	$5.003\times10^{+4}$	44.21	< .001	0.8468
	Greenhouse-Geisser	$1.501\times10^{+5}$	1.647	$9.111\times10^{+4}$	44.21	< .001	0.8468
Residuals	None	$2.716\times10^{+4}$	24.00	1132			
	Greenhouse-Geisser	$2.716\times10^{+4}$	13.18	2061			

Note. Type III Sum of Squares

Between Subjects Effects

Cases	Sum of Squares	df	Mean Square	F	p
Residuals	$8.208\times10^{+4}$	8	$1.026\times10^{+4}$		

Note. Type III Sum of Squares

Descriptives

Descriptives

Condition	N	Mean	SD	SE	Coefficient of variation
Free	9	720.2	87.21	29.07	0.1211
Simple	9	552.1	29.05	9.682	0.05262
Go/No-Go	9	575.1	51.40	17.13	0.08937
Go/Stop	9	607.4	50.65	16.88	0.08338

Assumption Checks

Test of Sphericity

	Mauchly's W	Approx. X^2	df	p-value	Greenhouse-Geisser ε	Huynh-Feldt ε	Lower Bound ε
Condition	0.2116	10.44	5	.066	0.5491	0.6730	0.3333

Post Hoc Tests

Post Hoc Comparisons - Condition

		Mean Difference	SE	df	t	Cohen's d	p_{Holm}
Free	Simple	168.2	22.74	8	7.397	2.878	< .001
	Go/No-Go	145.1	19.26	8	7.535	2.484	< .001
	Go/Stop	112.8	15.45	8	7.303	1.931	< .001
Simple	Go/No-Go	−23.07	10.96	8	−2.105	−0.3949	.068
	Go/Stop	−55.34	10.62	8	−5.213	−0.9472	.002
Go/No-Go	Go/Stop	−32.27	12.23	8	−2.639	−0.5523	.059

Note. P-value adjusted for comparing a family of 6 estimates.

Nonparametrics

Friedman Test

Factor	X^2_F	df	p	Kendall's W
Condition	23.80	3	< .001	0.8815

PeakVelTop_Repeated Measures ANOVA

Within Subjects Effects

Cases	Sphericity Correction	Sum of Squares	df	Mean Square	F	p	η^2
Condition	None	1.069	3.000	0.3562	1.601	.215	0.1668
	Greenhouse-Geisser	1.069	2.432	0.4393	1.601	.225	0.1668
Residuals	None	5.339	24.00	0.2224			
	Greenhouse-Geisser	5.339	19.46	0.2744			

Note. Type III Sum of Squares

Between Subjects Effects

Cases	Sum of Squares	df	Mean Square	F	p
Residuals	160.2	8	20.03		

Note. Type III Sum of Squares

Descriptives

Descriptives

Condition	N	Mean	SD	SE	Coefficient of variation
Free	9	31.63	2.348	0.7827	0.07423
Simple	9	31.50	2.077	0.6922	0.06592
Go/No-Go	9	31.75	2.316	0.7721	0.07295
Go/Stop	9	31.97	2.346	0.7821	0.07339

Assumption Checks

Test of Sphericity

	Mauchly's W	Approx. X^2	df	p-value	Greenhouse-Geisser ε	Huynh-Feldt ε	Lower Bound ε
Condition	0.6588	2.806	5	.732	0.8107	1.000	0.3333

Post Hoc Tests

Post Hoc Comparisons - Condition

		Mean Difference	SE	df	t	Cohen's d	p_{Holm}
Free	Simple	0.1273	0.2394	8	0.5317	0.05595	1.000
	Go/No-Go	−0.1220	0.2291	8	−0.5328	−0.05365	1.000
	Go/Stop	−0.3401	0.1893	8	−1.797	−0.1495	.550
Simple	Go/No-Go	−0.2493	0.2788	8	−0.8942	−0.1096	1.000
	Go/Stop	−0.4674	0.2115	8	−2.210	−0.2055	.349
Go/No-Go	Go/Stop	−0.2181	0.1689	8	−1.292	−0.09587	.930

Note. P-value adjusted for comparing a family of 6 estimates.

Nonparametrics

Friedman Test

Factor	X^2_F	df	p	Kendall's W
Condition	1.800	3	.615	0.06667

PeakFz_Repeated Measures ANOVA

Within Subjects Effects

Cases	Sphericity Correction	Sum of Squares	df	Mean Square	F	p	η^2
Condition	None	0.04398	3.000	0.01466	3.532	.030	0.3063
	Greenhouse-Geisser	0.04398	2.277	0.01931	3.532	.046	0.3063
Residuals	None	0.09960	24.00	0.004150			
	Greenhouse-Geisser	0.09960	18.22	0.005467			

Note. Type III Sum of Squares

Between Subjects Effects

Cases	Sum of Squares	df	Mean Square	F	p
Residuals	0.5673	8	0.07092		

Note. Type III Sum of Squares

Descriptives

Descriptives

Condition	N	Mean	SD	SE	Coefficient of variation
Free	9	1.316	0.1580	0.05265	0.1200
Simple	9	1.226	0.1346	0.04488	0.1098
Go/No-Go	9	1.236	0.1294	0.04313	0.1047
Go/Stop	9	1.265	0.1535	0.05115	0.1213

Assumption Checks

Test of Sphericity

	Mauchly's W	Approx. X^2	df	p-value	Greenhouse-Geisser ε	Huynh-Feldt ε	Lower Bound ε
Condition	0.5515	4.000	5	.553	0.7591	1.000	0.3333

Post Hoc Tests

Post Hoc Comparisons - Condition

		Mean Difference	SE	df	t	Cohen's d	p_{Holm}
Free	Simple	0.08992	0.03772	8	2.384	0.6228	.221
	Go/No-Go	0.07996	0.02719	8	2.940	0.5539	.112
	Go/Stop	0.05063	0.02928	8	1.729	0.3507	.488
Simple	Go/No-Go	−0.009956	0.02180	8	−0.4566	−0.06896	.886
	Go/Stop	−0.03929	0.03509	8	−1.120	−0.2722	.886
Go/No-Go	Go/Stop	−0.02933	0.02842	8	−1.032	−0.2032	.886

Note. P-value adjusted for comparing a family of 6 estimates.

Nonparametrics

Friedman Test

Factor	X^2_F	df	p	Kendall's W
Condition	6.200	3	.102	0.2296

HookTime_Repeated Measures ANOVA

Within Subjects Effects

Cases	Sphericity Correction	Sum of Squares	df	Mean Square	F	p	η^2
Condition	None	$1.341\times10^{+5}$	3.000	$4.470\times10^{+4}$	32.67	< .001	0.8033
	Greenhouse-Geisser	$1.341\times10^{+5}$	1.593	$8.417\times10^{+4}$	32.67	< .001	0.8033
Residuals	None	$3.283\times10^{+4}$	24.00	1368			
	Greenhouse-Geisser	$3.283\times10^{+4}$	12.75	2576			

Note. Type III Sum of Squares

Between Subjects Effects

Cases	Sum of Squares	df	Mean Square	F	p
Residuals	$1.959\times10^{+5}$	8	$2.448\times10^{+4}$		

Note. Type III Sum of Squares

Descriptives

Descriptives

Condition	N	Mean	SD	SE	Coefficient of variation
Free	9	421.8	107.9	35.98	0.2559
Simple	9	262.1	77.11	25.70	0.2942
Go/No-Go	9	288.7	69.44	23.15	0.2406
Go/Stop	9	342.9	78.54	26.18	0.2291

Assumption Checks

Test of Sphericity

	Mauchly's W	Approx. X^2	df	p-value	Greenhouse-Geisser ε	Huynh-Feldt ε	Lower Bound ε
Condition	0.2318	9.828	5	.083	0.5311	0.6420	0.3333

Post Hoc Tests

Post Hoc Comparisons - Condition

		Mean Difference	SE	df	t	Cohen's d	p_{Holm}
Free	Simple	159.7	24.58	8	6.496	1.889	< .001
	Go/No-Go	133.1	19.26	8	6.912	1.575	< .001
	Go/Stop	78.89	13.16	8	5.996	0.9332	.001
Simple	Go/No-Go	−26.55	10.77	8	−2.466	−0.3140	.039
	Go/Stop	−80.77	19.50	8	−4.142	−0.9554	.010
Go/No-Go	Go/Stop	−54.22	13.41	8	−4.043	−0.6414	.010

Note. P-value adjusted for comparing a family of 6 estimates.

Nonparametrics

Friedman Test

Factor	X^2_F	df	p	Kendall's W
Condition	25.93	3	< .001	0.9605

LoopTime_Repeated Measures ANOVA

Within Subjects Effects

Cases	Sphericity Correction	Sum of Squares	df	Mean Square	F	p	η^2
Condition	None	4203	3.000	1401	3.939	.020	0.3299
	Greenhouse-Geisser	4203	1.728	2433	3.939	.049	0.3299
Residuals	None	8535	24.00	355.6			
	Greenhouse-Geisser	8535	13.82	617.6			

Note. Type III Sum of Squares

Between Subjects Effects

Cases	Sum of Squares	df	Mean Square	F	p
Residuals	$6.674\times10^{+4}$	8	8343		

Note. Type III Sum of Squares

Descriptives

Descriptives

Condition	N	Mean	SD	SE	Coefficient of variation
Free	9	226.4	46.43	15.48	0.2051
Simple	9	221.6	59.51	19.84	0.2685
Go/No-Go	9	215.8	45.10	15.03	0.2090
Go/Stop	9	197.9	40.97	13.66	0.2070

Assumption Checks

Test of Sphericity

	Mauchly's W	Approx. X^2	df	p-value	Greenhouse-Geisser ε	Huynh-Feldt ε	Lower Bound ε
Condition	0.3308	7.437	5	.194	0.5758	0.7199	0.3333

Post Hoc Tests

Post Hoc Comparisons - Condition

		Mean Difference	SE	df	t	Cohen's d	p_{Holm}
Free	Simple	4.772	11.06	8	0.4314	0.09838	1.000
	Go/No-Go	10.65	5.536	8	1.923	0.2195	.363
	Go/Stop	28.52	6.056	8	4.710	0.5881	.009
Simple	Go/No-Go	5.876	9.018	8	0.6516	0.1212	1.000
	Go/Stop	23.75	12.41	8	1.913	0.4897	.363
Go/No-Go	Go/Stop	17.87	7.004	8	2.552	0.3685	.170

Note. P-value adjusted for comparing a family of 6 estimates.

Nonparametrics

Friedman Test

Factor	X^2_F	df	p	Kendall's W
Condition	8.600	3	.035	0.3185

AccelTime_Repeated Measures ANOVA

Within Subjects Effects

Cases	Sphericity Correction	Sum of Squares	df	Mean Square	F	p	η^2
Condition	None	151.4[a]	3.000[a]	50.47[a]	2.525[a]	.082[a]	0.2399
	Greenhouse-Geisser	151.4	1.759	86.06	2.525	.120	0.2399
Residuals	None	479.7	24.00	19.99			
	Greenhouse-Geisser	479.7	14.08	34.08			

Note. Type III Sum of Squares

[a] Mauchly's test of sphericity indicates that the assumption of sphericity is violated (p < .05).

Between Subjects Effects

Cases	Sum of Squares	df	Mean Square	F	p
Residuals	2282	8	285.2		

Note. Type III Sum of Squares

Descriptives

Descriptives

Condition	N	Mean	SD	SE	Coefficient of variation
Free	9	71.62	10.06	3.353	0.1404
Simple	9	67.72	10.43	3.478	0.1541
Go/No-Go	9	70.16	8.492	2.831	0.1210
Go/Stop	9	66.36	7.938	2.646	0.1196

Assumption Checks

Test of Sphericity

	Mauchly's W	Approx. X^2	df	p-value	Greenhouse-Geisser ε	Huynh-Feldt ε	Lower Bound ε
Condition	0.1615	12.25	5	.033	0.5865	0.7390	0.3333

Post Hoc Tests

Post Hoc Comparisons - Condition

		Mean Difference	SE	df	t	Cohen's d	p_{Holm}
Free	Simple	3.907	2.712	8	1.441	0.4206	.563
	Go/No-Go	1.466	0.8693	8	1.687	0.1578	.521
	Go/Stop	5.261	1.849	8	2.846	0.5664	.129
Simple	Go/No-Go	−2.441	2.560	8	−0.9535	−0.2628	.737
	Go/Stop	1.354	2.605	8	0.5197	0.1458	.737
Go/No-Go	Go/Stop	3.795	1.333	8	2.848	0.4086	.129

Note. P-value adjusted for comparing a family of 6 estimates.

Nonparametrics

Friedman Test

Factor	X^2_F	df	p	Kendall's W
Condition	9.000	3	.029	0.3333